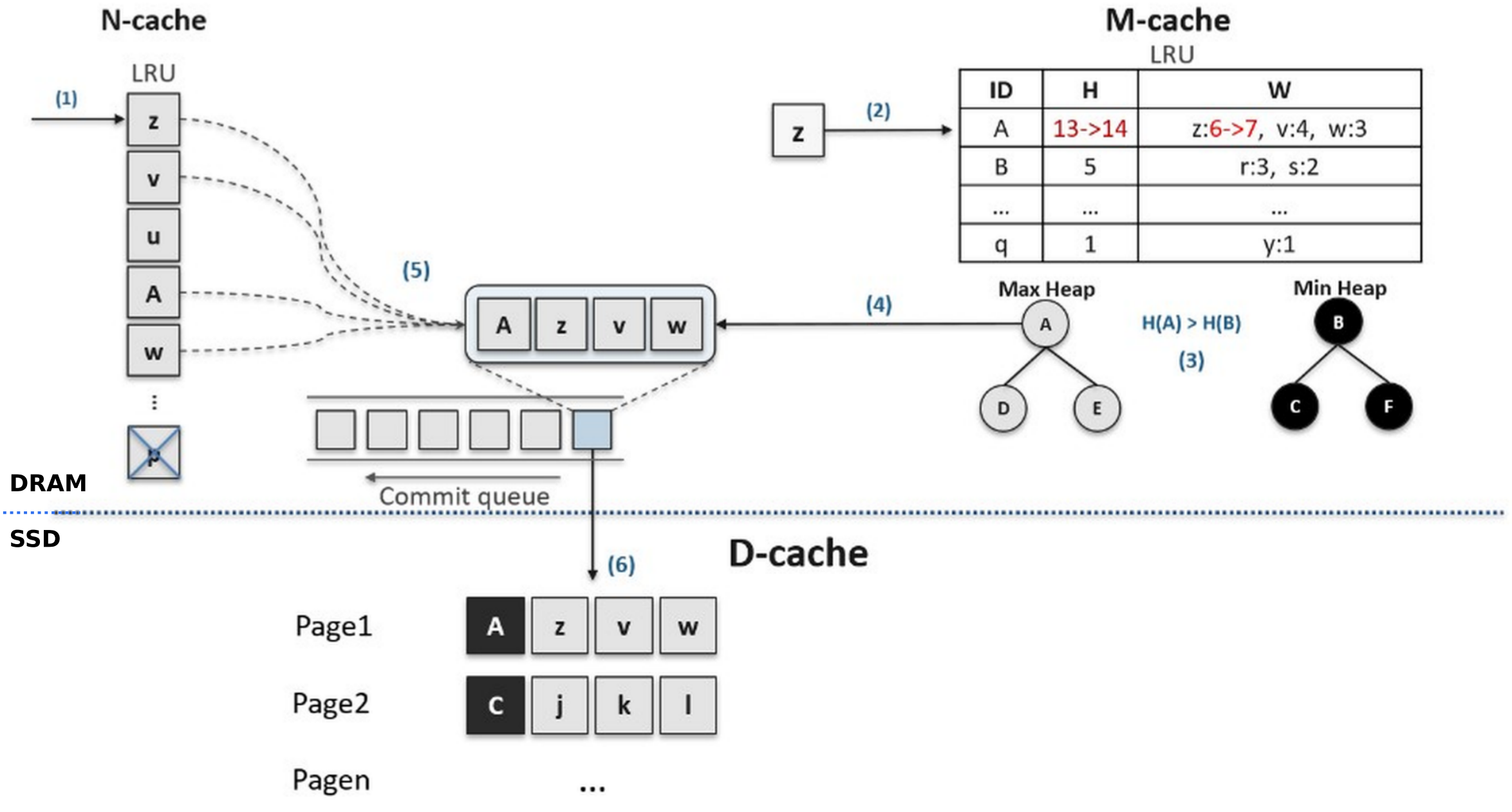

N-cache
LRU
(1)
z
v
u
A
w
p
M-cache
LRU
z
(2)
ID
H
W
A
13->14
z:6->7, v:4, w:3
B
5
r:3, s:2
...
...
...
q
1
y:1
(5)
A
z
v
w
(4)
Max Heap
A
D
E
H(A) > H(B)
(3)
Min Heap
B
C
F
Commit queue
DRAM
SSD
D-cache
(6)
Page1
A
z
v
w
Page2
C
j
k
l
Pagen
...